SQL · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Query order

- Written order: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, LIMIT.
- Logical run order: FROM and JOIN, WHERE, GROUP BY, HAVING, SELECT, DISTINCT, ORDER BY, LIMIT.
- That is why a column alias from SELECT cannot be used in WHERE, but can be used in ORDER BY.

Filtering and NULL

- WHERE filters rows; HAVING filters groups after aggregation.
- NULL means unknown: use IS NULL and IS NOT NULL, never = NULL.
- COUNT(*) counts all rows; COUNT(col) skips NULLs; AVG, SUM, MIN and MAX ignore NULLs.
- COALESCE(col, 'N/A') returns the first non-NULL value.
- NOT IN with a NULL in the list returns no rows; use NOT EXISTS instead.

Joins

- INNER: only matching rows. LEFT: all left rows plus matches, NULLs where none.
- FULL OUTER: all rows from both sides. CROSS: every combination (Cartesian product).
- Anti-join to find missing rows: LEFT JOIN b ... WHERE b.id IS NULL.
- Self join: a table joined to itself, such as employee and manager.

Set operations and subqueries

- UNION removes duplicates; UNION ALL keeps them and is faster.
- INTERSECT gives common rows; EXCEPT (MINUS in Oracle) gives the difference.
- Correlated subquery: runs once per outer row because it uses an outer column.
- CTE: WITH name AS (...) for readable steps; recursive CTEs walk hierarchies.

Window functions

- ROW_NUMBER: 1, 2, 3 with no ties. RANK: 1, 2, 2, 4. DENSE_RANK: 1, 2, 2, 3.
- PARTITION BY restarts per group but keeps every row (unlike GROUP BY).
- LAG and LEAD read the previous and next row. SUM(x) OVER (ORDER BY d) gives a running total.
- Second highest salary: DENSE_RANK() OVER (ORDER BY salary DESC) = 2.
- Delete duplicates: ROW_NUMBER() OVER (PARTITION BY dup columns) > 1.

Keys, constraints and normalization

- PRIMARY KEY: unique and not NULL. FOREIGN KEY: references another table's key.
- UNIQUE, NOT NULL, CHECK and DEFAULT constraints protect data quality.
- 1NF atomic values; 2NF no partial dependency; 3NF no transitive dependency.
- DELETE (filtered, row-by-row) vs TRUNCATE (all rows, fast) vs DROP (removes the table).

Indexes and performance

- Clustered index sets the physical row order (one per table); non-clustered indexes are separate.
- Composite index (a, b) helps filters on a, or on a and b.
- Covering index holds every column a query needs.
- Avoid functions on indexed columns in WHERE (not sargable). Avoid SELECT *.
- Read the execution plan: look for full scans, missing indexes and bad join types.

Transactions

- ACID: Atomicity, Consistency, Isolation, Durability.
- COMMIT saves, ROLLBACK undoes, SAVEPOINT marks a point to roll back to.
- Isolation levels: READ UNCOMMITTED (dirty reads), READ COMMITTED, REPEATABLE READ, SERIALIZABLE (no phantoms).
- Deadlock: two transactions wait on each other; the database kills one.

Support tips

- Always SELECT with the same WHERE before running UPDATE or DELETE.
- Use parameterized queries to prevent SQL injection.
- Reconcile data with counts and sums per stage to find where it diverges.